HAT-P-32b

R-0210

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_241228` · created 2026-07-17T00:08:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460672.6728 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.014
Transit depth (Rp/Rs)^2	1.83 +/- 0.37 [%]
Orbital Inclination (inc)	84.44 +/- 1.37
Airmass coefficient 1 (a1)	1.1528 +/- 0.0075
Airmass coefficient 2 (a2)	-0.111 +/- 0.019
Scatter in the residuals of the lightcurve fit is	0.66 %
Best Comparison Star	None
Optimal Aperture	4.77
Optimal Annulus	5.86
Transit Duration (day)	0.1125 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.014 vs 0.1489 (deviation 0.87σ)
Duration fit vs published	0.1125 d vs 0.1304 d (deviation 3.64σ)
Mid-time O–C	3.0 min
Depth SNR	8.4
Residual scatter	0.66 %

Light curve

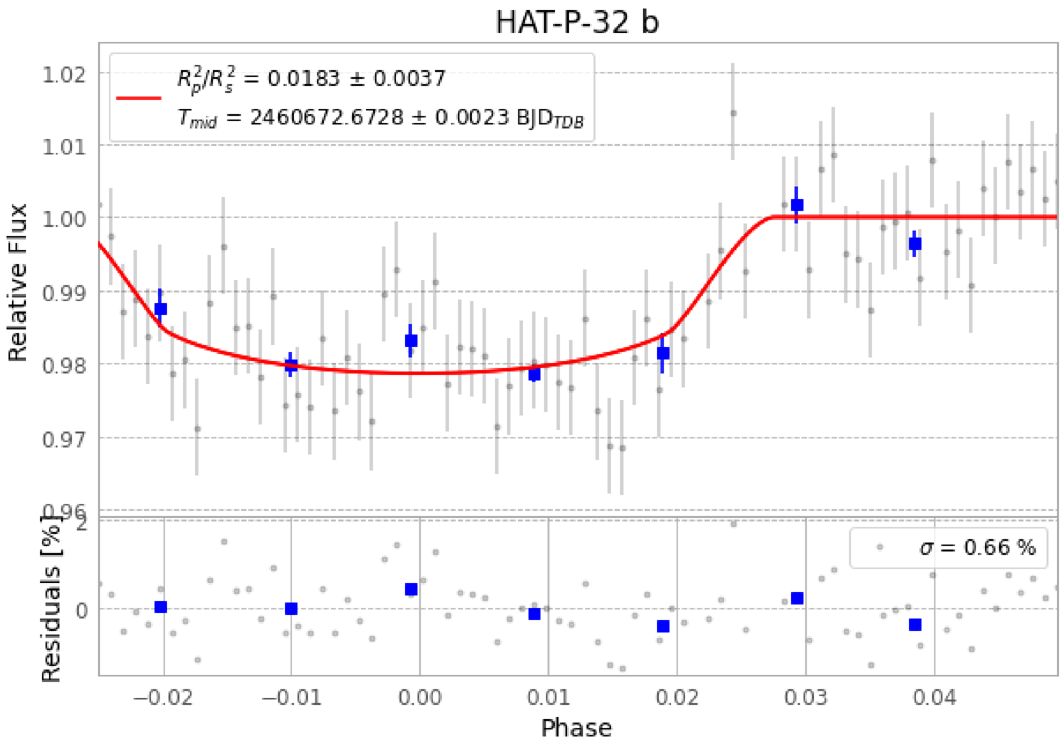

Figure 1 — FinalLightCurve_HAT-P-32 b_2024-12-27.png (SHA-256 84d462ebf71655ef…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 266], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e51cecd6ba29b07d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e`

Data provenance

78 FITS frames (51 MB), `HATP-32241228024515.FITS` → `HATP-32241228063615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241228.log (1f2812e223dd…), FinalLightCurve_HAT-P-32 b_2024-12-27.png (84d462ebf716…), FinalLightCurve_HAT-P-32 b_2024-12-27.csv (3c9624b80829…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

